

BROCKVILLE, ON, Canada

WMO: 711610

Lat: 44.6354N

Lon: 75.7516W

Elev: 120

StdP: 99.89

Time Zone: -5.00 (NAE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-24.5	-21.1	-28.6	0.3	-23.9	-25.2	0.4	-20.4	11.2	-3.7	9.8	-3.0	2.5	310	0.541

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.9	29.7	21.6	28.2	20.7	26.9	20.2	23.2	27.8	22.3	26.5	21.4	25.2	4.4	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.7	16.6	26.2	20.9	15.7	24.8	20.0	15.0	23.9	69.8	27.9	66.1	26.6	62.9	25.3	27.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.0	7.9	6.9	DB	-30.5	32.3	3.0	1.7	-32.6	33.6	-34.4	34.6	-36.1	35.6	-38.2	36.9	(4)
(5)			WB	-30.6	25.2	3.0	1.0	-32.7	25.9	-34.5	26.5	-36.1	27.1	-38.3	27.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	7.2	-8.2	-6.6	-1.2	6.4	13.9	17.7	20.8	19.8	15.7	9.2	2.3	-4.1
	DBStd	11.30	7.08	6.50	6.24	4.66	4.45	3.31	2.90	2.77	4.22	4.63	5.31	6.60
	HDD10.0	2268	565	464	351	127	14	0	0	0	3	72	234	436
	HDD18.3	4273	823	698	606	359	149	51	9	17	101	284	480	694
	CDD10.0	1252	0	0	3	19	136	230	335	303	175	47	4	0
	CDD18.3	214	0	0	0	1	12	31	86	62	22	1	0	0
	CDH23.3	1669	0	0	1	14	126	233	704	412	175	4	0	0
	CDH26.7	346	0	0	0	2	23	36	184	73	29	0	0	0
Wind	WSAvg	3.2	3.5	3.6	3.6	3.8	3.2	2.9	2.7	2.6	2.7	3.2	3.3	3.2
Precipitation	PrecAvg	984	75	62	75	81	77	74	79	92	92	82	102	91
	PrecMax	1219	149	145	124	149	122	136	197	181	189	137	166	149
	PrecMin	758	26	14	29	28	31	14	34	24	27	30	50	39
	PrecStd	102	33	34	28	33	30	30	39	35	37	32	29	24
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.2	10.7	20.8	25.8	29.1	29.7	32.4	30.7	29.4	23.9	18.4	12.9
		MCWB	9.2	9.0	15.1	17.2	20.5	21.7	23.2	21.2	22.1	18.6	13.0	10.2
	2%	DB	6.5	6.9	13.2	20.1	26.6	27.6	30.5	28.5	27.2	21.3	14.9	8.9
		MCWB	4.8	5.0	8.7	13.0	18.6	19.9	21.9	21.2	21.2	16.8	10.8	7.7
	5%	DB	3.5	4.6	9.8	16.9	24.4	26.0	28.7	27.1	25.4	18.9	12.5	6.8
		MCWB	2.1	2.6	6.2	11.1	17.2	18.7	21.0	20.6	19.9	14.5	9.4	5.8
	10%	DB	1.4	2.4	7.0	14.3	22.1	24.4	27.2	25.8	23.4	16.3	10.4	4.1
		MCWB	0.3	0.9	3.8	9.0	15.8	17.9	20.4	19.9	18.9	13.1	7.8	2.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.4	9.0	15.1	17.4	21.9	23.3	25.1	23.7	23.5	19.9	13.6	11.2
		MCDB	10.3	10.2	20.6	23.8	27.7	28.1	30.7	28.3	27.9	22.4	16.6	12.1
	2%	WB	5.3	5.2	9.3	14.3	20.0	21.7	23.6	22.7	22.2	17.5	11.7	7.8
		MCDB	6.0	7.0	12.5	18.4	25.0	26.1	28.3	26.9	26.1	20.4	14.5	8.8
	5%	WB	2.2	2.7	6.6	11.7	18.5	20.3	22.5	21.7	20.8	15.3	9.8	5.6
		MCDB	3.3	4.1	9.4	15.6	22.8	23.8	26.7	25.5	24.0	18.1	12.1	6.9
	10%	WB	0.5	1.0	4.2	9.8	16.9	19.1	21.5	20.8	19.6	13.6	7.9	3.0
		MCDB	1.4	2.3	6.5	13.8	20.3	22.5	25.3	24.1	22.5	15.8	10.4	4.0
Mean Daily Temperature Range	5% DB	MDBR	9.4	9.3	9.2	10.2	11.2	10.3	10.9	10.6	11.2	9.1	8.4	7.6
		MCDBR	9.8	10.2	12.3	15.1	14.3	12.8	12.8	12.1	12.7	11.5	11.9	9.2
		MCWBR	9.3	9.0	8.4	9.2	7.8	6.7	6.1	6.1	7.2	7.5	8.9	8.6
	5% WB	MCDBR	9.6	10.3	11.1	13.1	11.9	9.9	10.6	9.8	11.3	10.6	10.6	9.5
MCWBR		9.3	9.4	8.4	9.4	7.6	5.9	5.8	5.7	7.1	7.6	8.6	9.0	
Clear-Sky Solar Irradiance	taub	0.285	0.292	0.320	0.349	0.379	0.413	0.423	0.412	0.365	0.344	0.306	0.281	
	taud	2.379	2.362	2.422	2.405	2.363	2.280	2.256	2.292	2.411	2.466	2.511	2.449	
	Ebn at Noon	836	899	918	913	891	858	843	840	857	828	811	808	
	Edh at Noon	75	92	100	111	119	130	132	123	101	83	67	64	
All-Sky Solar Radiation	RadAvg	1.36	2.26	3.58	4.42	5.26	5.64	5.78	5.09	4.04	2.39	1.50	1.03	
	RadStd	0.14	0.23	0.29	0.53	0.54	0.45	0.49	0.23	0.34	0.23	0.16	0.12	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (33 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page